

MATH 20250 HW2: SAMPLES SOLUTIONS AND MISTAKES

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1. SAMPLES

1.1. Problem 7.1.

Proof. Let $x, y \in X \cap Y$. Thus, $x + y \in X$ and $x + y \in Y$ and thus $x + y \in X \cap Y$.

Similarly, given $x \in X \cap Y$, and $a \in \mathbb{R}$, there is $ax \in X$ and $ax \in Y$ and thus $ax \in X \cap Y$.

Thus, $X \cap Y$ is a subspace. $\square$

1.2. Problem 7.2.

Proof. Let $u, v \in X + Y$. Then $\exists x_1, x_2 \in X$ and $y_1, y_2 \in Y$ such that $u = x_1 + y_1, v = x_2 + y_2$. Then $u + v = (x_1 + x_2) + (y_1 + y_2)$. Since $x_1 + x_2 \in X, y_1 + y_2 \in Y$, $u + v \in X + Y$.

Similarly, given $a \in \mathbb{R}$, $au = ax_1 + ay_1 \in X + Y$.

Thus $X + Y$ is a subspace. $\square$

1.3. Problem 7.3.

Proof. Assume for contradiction that $x + v \in X$. Then, $(x + v) - x = v \in X$, contradict that $v \notin X$.

Thus, $x + v \notin X$. $\square$

1.4. Problem 7.4.

Proof. First prove that $X \cup Y$ is a subspace $\longrightarrow X \subseteq Y$ or $Y \subseteq X$.

We will prove the contrapositive. Assume that neither $X \subseteq Y$ nor $Y \subseteq X$ is true, i.e., $\exists u \in X - Y$ and $v \in Y - X$.

By (7.3), since $u \in X, v \notin X, u+v \notin X$. Similarly, $u+v \notin Y$. Thus, $x+y \notin X \cup Y$. Thus, $X \cup Y$ is not a subspace.

Then we shall prove that $X \subseteq Y$ or $Y \subseteq X \longrightarrow X \cup Y$ is a subspace. If $X \subseteq Y$, then $X \cup Y = Y$; and if $Y \subseteq X$, then $X \cup Y = X$. In both cases $X \cup Y$ is a subspace. $\square$

2. NOTES

2.1. **Q2.3.4.** Again, $\mathbb{P}_3$ is the set of polynomials with at most degree 3, i.e. polynomials of the form $a_0 + a_1x + a_2x^2 + a_3x^3$. Thus, it is generated by a basis $\{1, x, x^2, x^3\}$. Thus, $\mathbb{P}_3$ is four-dimensional, so it cannot be generated by three polynomials.

2.2. **About Elementary Row Transformation.** Some students use $=$ to denote elementary row transformation of matrices. This is of course non-standard. Use a right arrow $\Longrightarrow$ instead. In LaTeX this is `\Longrightarrow`.

2.3. **About LADW Theorem 2.3.1.** The LADW Theorem 2.3.1 states that for vectors $v_1, \dots, v_n$, if the matrix $A = [v_1, \dots, v_n]$ after reduced to the row-echelon form has a pivot on every row, then $v_1, \dots, v_n$ span the entire space, and vice versa. However, it has to be noticed that to apply LADW Theorem 2.3.1, the vectors would have to be written as column vectors. For example, when there are $v_1, v_2, v_3 \in \mathbb{R}^4$, $[v_1, \dots, v_n]$ would be a 4×3 matrix (having 4 rows and 3 columns).

Errors are likely to occur if the student mistakenly let $v_1, \dots, v_n$ be the rows of the matrix A . For example, when $n = 3$ and $v_1, v_2, v_3 \in \mathbb{R}^4$, then A would be 3×4 matrix. If the row-echelon form A happens to have a pivot at on every row, then the students who attempt to apply LADW Theorem 2.3.1 would conclude that three vectors v_1, v_2, v_3 can span a four-dimensional space such as $\mathbb{R}^4$, which is entirely impossible.

2.4. **Comment about Quoting Theorems.** Using a Linear Algebra theorem that is not yet introduced in class is not recommended, since we are probably still going through results that lead up to the theorem.

In future study, quoting a theorem is OK and recommended if you are quoting the result from a different branch of study. For example, you can quote results from calculus in this course.